FISCAL AND MONETARY POLICY INTERACTIONS

Fiscal and Monetary Policy 2026

Piotr Żoch

May 25, 2026

ROADMAP

- Fixed-regime benchmark: root counts and shock responses.
- Regime switching: how future regimes change today's coefficients.
- Identification: why equilibrium correlations do not reveal structural regimes.
- Valuation: why low rates and default require extra structure.
- Fiscal compatibility: how policy rules restrict the target price path.

HOW THE PIECES FIT

We start from one benchmark and then change one ingredient at a time.

- The first model is the linear Fisher–budget system; it gives root counts and shock coefficients.
- Switching keeps the same logic but lets policy coefficients follow a Markov chain.
- Identification asks what an econometrician sees from equilibrium data.
- Low rates and default change the valuation equation.
- Fiscal compatibility asks whether a target price path has enough fiscal backing.

FIXED-REGIME BENCHMARK: ROOT-COUNT DETERMINACY

	Fiscal passive	Fiscal active
Monetary active	Regime M	AM/AF: no eq.
Monetary passive	PM/PF: indeterminate	Regime F

- One active + one passive $\Rightarrow$ determinate equilibrium.
- Other combinations create no bounded equilibrium or indeterminacy in this benchmark.
- These statements assume the regime is fixed forever.

THE FIXED-REGIME SYSTEM

Use level deviations of gross nominal objects, not log deviations:

$$i_t \equiv R_t - \bar{R}, \qquad \pi_t \equiv \Pi_t - \bar{\Pi}, \qquad \bar{\Pi} = 1, \quad \bar{R} = \beta^{-1}.$$

Linearizing Fisher gives $\mathbb{E}_t \pi_{t+1} = \beta i_t$. The policy rules and normalized budget constraint are

$$\begin{aligned} i_t &= \phi_\pi \pi_t + \theta_t, \\ \tau_t &= \gamma b_{t-1} + \psi_t, \\ b_t &= \beta^{-1} b_{t-1} - \beta^{-1} \pi_t - \tau_t. \end{aligned}$$

- b_t, τ_t are normalized so the inflation-revaluation coefficient is β^{-1}.
- θ_t is a monetary-rule intercept shock; $\psi_t > 0$ is a surplus shock.
- Monetary active means $\beta\phi_\pi > 1$.

MATRIX FORM

Combining the policy rules with Fisher and the government budget constraint:

$$\begin{bmatrix} \mathbb{E}_t \pi_{t+1} \\ b_t \end{bmatrix} = \underbrace{\begin{bmatrix} \phi_\pi \beta & 0 \\ -\beta^{-1} & \beta^{-1} - \gamma \end{bmatrix}}_{A} \begin{bmatrix} \pi_t \\ b_{t-1} \end{bmatrix} + \begin{bmatrix} \beta \theta_t \\ -\psi_t \end{bmatrix}.$$

- π_t is a jump variable.
- b_{t-1} is a predetermined state.
- Blanchard–Kahn conditions require exactly one unstable root.

EIGENVALUE ALGEBRA

The matrix is triangular, so:

$$
\begin{aligned}
\det(A - \lambda I) &= (\phi_\pi \beta - \lambda)(\beta^{-1} - \gamma - \lambda), \\
\lambda_\pi &= \phi_\pi \beta, \\
\lambda_b &= \beta^{-1} - \gamma.
\end{aligned}
$$

- Monetary active: $\lambda_\pi > 1 \Leftrightarrow \phi_\pi > 1/\beta$.
- Monetary passive: $\lambda_\pi < 1$.
- Fiscal passive: $|\lambda_b| < 1$.
- Fiscal active, economically relevant positive-root case: $\lambda_b > 1 \Leftrightarrow \gamma < \beta^{-1} - 1$.
- Explosive negative-root fiscal cases are excluded from the simple taxonomy below.

REGIMES ARE ROOT COUNTS

Regime	Roots	Unstable roots	Outcome
AM/PF	$\lambda_\pi > 1,\ \lvert\lambda_b\rvert < 1$	one	Regime M
PM/AF	$\lambda_\pi < 1,\ \lambda_b > 1$	one	Regime F
AM/AF	$\lambda_\pi > 1,\ \lambda_b > 1$	two	No bounded eq.
PM/PF	$\lambda_\pi < 1,\ \lvert\lambda_b\rvert < 1$	zero	Indeterminate

- Regime M: boundedness selects inflation; fiscal policy stabilizes debt.
- Regime F: the price level jumps to put inherited real debt on the fiscal valuation path.
- Root count is a local linear result, not a full historical classification.

FIXED-REGIME SOLUTION: SHOCK PROCESSES

Define

$$\lambda_\pi \equiv \phi_\pi \beta, \qquad \lambda_b \equiv \beta^{-1} - \gamma.$$

Let policy shocks follow

$$\theta_t = \rho_\theta \theta_{t-1} + \varepsilon_t^\theta, \qquad \psi_t = \rho_\psi \psi_{t-1} + \varepsilon_t^\psi, \qquad |\rho_\theta|, |\rho_\psi| < 1.$$

Guess a bounded linear solution:

$$\pi_t = h\, b_{t-1} + g^\theta \theta_t + g^\psi \psi_t.$$

FIXED-REGIME SOLUTION: DEBT LAW

The debt law implied by the guess is

$$b_t = (\lambda_b - \beta^{-1} h) b_{t-1} - \beta^{-1} g^{\theta} \theta_t - (1 + \beta^{-1} g^{\psi}) \psi_t.$$

The coefficient on inherited debt is

$$\lambda_b - \beta^{-1} h.$$

- In Regime M this root must be stable directly.
- In Regime F the price level jumps so this root becomes the stable monetary root.

FIXED-REGIME SOLUTION: COEFFICIENT EQUATIONS

Matching coefficients in the Fisher equation gives

$$
\begin{aligned}
h(\lambda_b - \beta^{-1}h) &= \lambda_\pi h, \\
g^\theta(\rho_\theta - \lambda_\pi - \beta^{-1}h) &= \beta, \\
g^\psi(\rho_\psi - \lambda_\pi - \beta^{-1}h) &= h.
\end{aligned}
$$

- The first equation has two roots:

$$h = 0 \qquad \text{or} \qquad h = \beta(\lambda_b - \lambda_\pi).$$

- Boundedness selects which root is the equilibrium root.

REGIME M SOLUTION: COEFFICIENTS

In Regime M,

$$\lambda_\pi > 1, \qquad |\lambda_b| < 1.$$

Bounded inflation selects

$$h_M = 0, \qquad g_M^\theta = -\frac{\beta}{\lambda_\pi - \rho_\theta}, \qquad \boxed{g_M^\psi = 0}.$$

- The zero fiscal-shock coefficient is the algebraic insulation result.

REGIME M SOLUTION: RESPONSES

The AM/PF solution is

$$\pi_t = -\frac{\beta}{\lambda_\pi - \rho_\theta}\theta_t,$$

and

$$b_t = \lambda_b b_{t-1} + \frac{1}{\lambda_\pi - \rho_\theta}\theta_t - \psi_t.$$

The nominal-rate response is

$$i_t = \phi_\pi \pi_t + \theta_t = -\frac{\rho_\theta}{\lambda_\pi - \rho_\theta}\theta_t.$$

With $\rho_\theta > 0$, a rate-increase IRF therefore uses $\theta_0 < 0$.

REGIME M: FISCAL SHOCKS ARE INSULATED

Set $\theta_t = 0$ and consider a fiscal shock ψ_t:

$$\pi_t = 0, \qquad i_t = 0, \qquad b_t = \lambda_b b_{t-1} - \psi_t.$$

Since fiscal policy is passive,

$$|\lambda_b| < 1,$$

debt converges back after the shock.

- A tax cut is $-\psi_t$: it raises debt, not inflation.
- Active monetary policy pins down inflation; passive fiscal policy absorbs the fiscal shock through future surpluses.
- This is the fixed-regime benchmark that switching will overturn.

REGIME F SOLUTION: PM/AF

In Regime F,

$$\lambda_\pi < 1, \qquad \lambda_b > 1.$$

Bounded debt selects

$$h_F = \beta(\lambda_b - \lambda_\pi).$$

The shock coefficients are

$$g_F^\theta = -\frac{\beta}{\lambda_b - \rho_\theta}, \qquad g_F^\psi = -\frac{\beta(\lambda_b - \lambda_\pi)}{\lambda_b - \rho_\psi}.$$

REGIME F: FISCAL SHOCKS MOVE INFLATION

The Regime F solution is

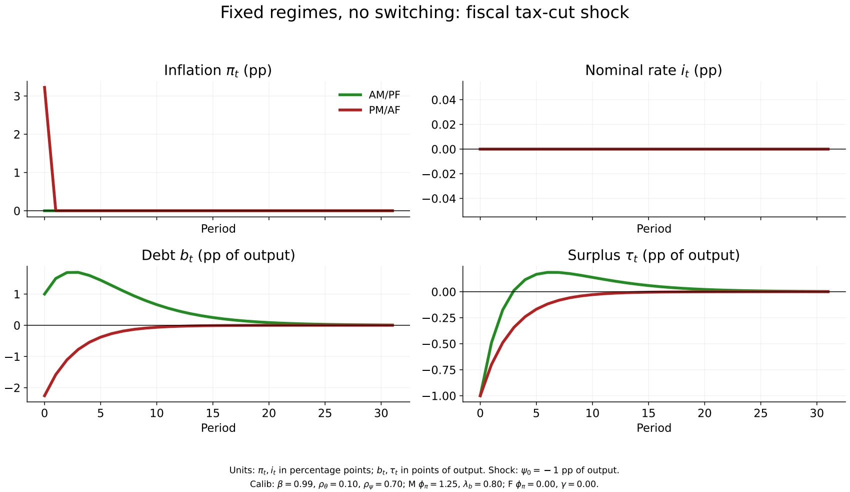
$$\pi_t = \beta(\lambda_b - \lambda_\pi) b_{t-1} - \frac{\beta}{\lambda_b - \rho_\theta}\theta_t - \frac{\beta(\lambda_b - \lambda_\pi)}{\lambda_b - \rho_\psi}\psi_t.$$

Debt evolves according to

$$b_t = \lambda_\pi b_{t-1} + \frac{1}{\lambda_b - \rho_\theta}\theta_t - \frac{\lambda_\pi - \rho_\psi}{\lambda_b - \rho_\psi}\psi_t.$$

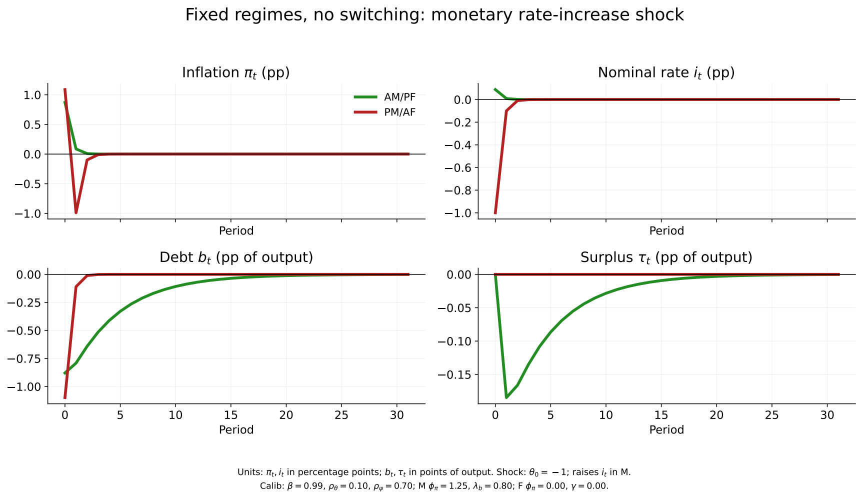
- The fiscal shock coefficient in inflation is nonzero.
- Active fiscal policy makes the price level revalue inherited nominal debt.

IRF: FISCAL SHOCK WITH FIXED REGIMES



No switching: AM/PF and PM/AF are fixed forever. Units: π_t, i_t in pp; b_t, τ_t in pp of output. Shock: $\psi_0 = -1$.

IRF: MONETARY RATE-INCREASE SHOCK WITH FIXED REGIMES



No switching: AM/PF and PM/AF are fixed forever. Units: π_t, i_t in pp; b_t, τ_t in pp of output. Shock: $\theta_0 = -1$; this raises i_t in Regime M.

MOTIVATION: POLICY-RULE COEFFICIENTS ARE TIME-VARYING

- The fixed-regime solution assumed ϕ_π and γ are constant forever.
- 1942–1951: Fed pegged 3-month Treasury bills at 3/8% to support war finance.
- 1965–1979: empirical Taylor coefficient fails the Taylor principle.
- 1980–2007: Volcker era. Active monetary, mostly passive fiscal.
- 2008–2015, 2020–2022: ZLB / near-ZLB episodes; debt/GDP rises sharply.
- 2022–2026: high-rate anti-inflation regime; fiscal regime still disputed.

WHAT CHANGES AFTER THE FIXED BENCHMARK?

The fixed-regime solution gave two clean results:

$$g_M^{\psi} = 0, \qquad g_F^{\psi} \neq 0.$$

The rest of the lecture asks what survives when we relax one assumption at a time.

- Regime switching: policy coefficients are stochastic, so today's coefficients depend on transition probabilities.
- Identification: equilibrium correlations mix policy rules with valuation equations.
- Low rates and default: the simple valuation equation needs extra structure.
- Fiscal compatibility: an inflation target requires fiscal backing of the target price path.

WHY FIXED REGIMES ARE INSUFFICIENT

- The root-count result is conditional on policy coefficients being fixed forever.
- Historical policy rules are time-varying: wartime pegs, Volcker disinflation, ZLB episodes, and post-2022 tightening have different monetary-fiscal environments.
- Technical question: what replaces the single matrix A when the policy-rule coefficients follow a Markov chain?
- Answer: solve for regime-contingent coefficients linked by the transition matrix.

REGIME SWITCHING

MARKOV-SWITCHING POLICY RULES

$$i_t = \bar{i}(\sigma_t) + \phi_\pi(\sigma_t)\,\pi_t + \theta_t,$$

$$\tau_t = \gamma_0(\sigma_t) + \gamma_1(\sigma_t)\,b_{t-1} + \psi_t,$$

where $\sigma_t \in \{M, F\}$ follows

$$\Pr(\sigma_{t+1} = j \mid \sigma_t = i) = p_{ij}, \qquad \Pi = \begin{bmatrix} p_{MM} & p_{MF} \\ p_{FM} & p_{FF} \end{bmatrix}.$$

- Regime M: AM/PF, $\phi_\pi(M) > 1/\beta$ and $|\beta^{-1} - \gamma_1(M)| < 1$.
- Regime F: PM/AF, $\phi_\pi(F) < 1/\beta$ and $\beta^{-1} - \gamma_1(F) > 1$ in the positive-root case.
- Private agents observe σ_t and know Π.
- Below write $\phi_{\pi,i} \equiv \phi_\pi(i)$ and $\gamma_i \equiv \gamma_1(i)$.

REGIME-CONTINGENT SOLUTION FORM

Assume a bounded linear solution:

$$\pi_t = h_{\sigma_t} b_{t-1} + g^{\theta}_{\sigma_t} \theta_t + g^{\psi}_{\sigma_t} \psi_t.$$

The expectation in the Fisher equation is now a cross-regime object:

$$\mathbb{E}_t \pi_{t+1} = \sum_{j \in \{M,F\}} p_{\sigma_t j} \, \mathbb{E}_t \left[h_j b_t + g^{\theta}_j \theta_{t+1} + g^{\psi}_j \psi_{t+1} \right].$$

- The coefficients $(h_i, g^{\theta}_i, g^{\psi}_i)$ are solved jointly across regimes.
- Today's inflation response depends on Π, not only on today's policy coefficients.

SWITCHING ALGEBRA: DEBT LAW OF MOTION

Suppress intercepts and shocks first. In state $i \in \{M, F\}$, conjecture

$$\pi_t = h_i b_{t-1}.$$

The government budget equation implies

$$b_t = \left(\beta^{-1} - \gamma_i - \beta^{-1} h_i\right) b_{t-1} \equiv q_i(h_i) b_{t-1}.$$

Thus $q_i(h_i)$ is the regime-i debt-transition coefficient:

$$q_i(h_i) = \beta^{-1} - \gamma_i - \beta^{-1} h_i.$$

SWITCHING ALGEBRA: DEBT COEFFICIENT EQUATION

The Fisher equation gives

$$\beta\phi_{\pi,i}h_i b_{t-1} = \sum_{j\in\{M,F\}} p_{ij}h_j b_t = \left(\sum_j p_{ij}h_j\right) q_i(h_i) b_{t-1}.$$

Therefore the two regime coefficients solve

$$\boxed{\beta\phi_{\pi,i}h_i = \left(\sum_j p_{ij}h_j\right)\left(\beta^{-1} - \gamma_i - \beta^{-1}h_i\right)}, \qquad i = M, F.$$

SWITCHING ALGEBRA: FISCAL-SHOCK SETUP

Let $H_i \equiv \sum_j p_{ij} h_j$ and let ψ_t be an i.i.d. surplus shock. Conjecture

$$\pi_t = h_i b_{t-1} + g_i^{\psi} \psi_t.$$

The coefficient on ψ_t in next period's debt is

$$\frac{\partial b_t}{\partial \psi_t} = -\left(1 + \beta^{-1} g_i^{\psi}\right).$$

Matching ψ_t coefficients in Fisher:

$$\beta \phi_{\pi,i} g_i^{\psi} = -H_i \left(1 + \beta^{-1} g_i^{\psi}\right), \qquad i = M, F.$$

SWITCHING ALGEBRA: FISCAL-SHOCK RESULT

Solving the coefficient equation gives

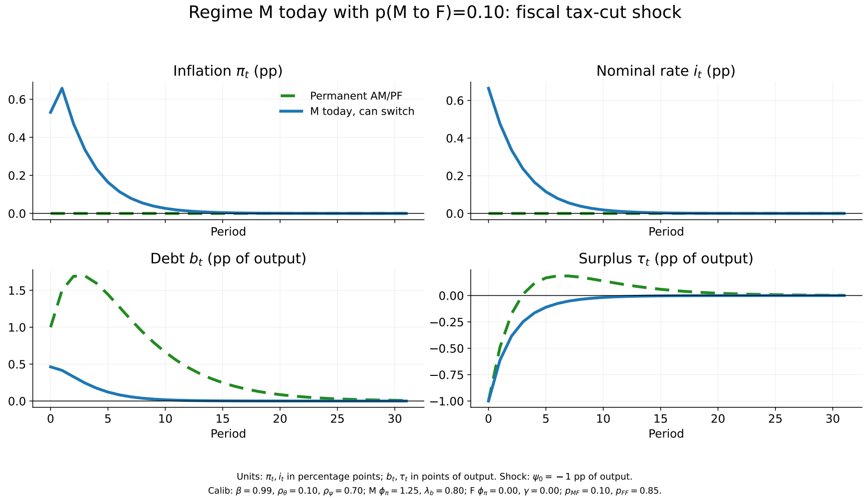
$$g_i^{\psi}\left(\beta\phi_{\pi,i} + \beta^{-1}H_i\right) = -H_i, \qquad H_i \equiv \sum_j p_{ij}h_j,$$

so

$$\boxed{g_i^{\psi} = -\frac{H_i}{\beta\phi_{\pi,i} + \beta^{-1}H_i}}.$$

- Permanent AM/PF: $p_{MF} = 0$ and $h_M = 0$ imply $H_M = 0$ and $g_M^{\psi} = 0$.
- Switching with $p_{MF} > 0$ and $h_F \neq 0$ generically makes $H_M \neq 0$, so $g_M^{\psi} \neq 0$.
- A tax cut is $-\psi_t$, so the inflation response has the opposite sign of g_i^{ψ}.

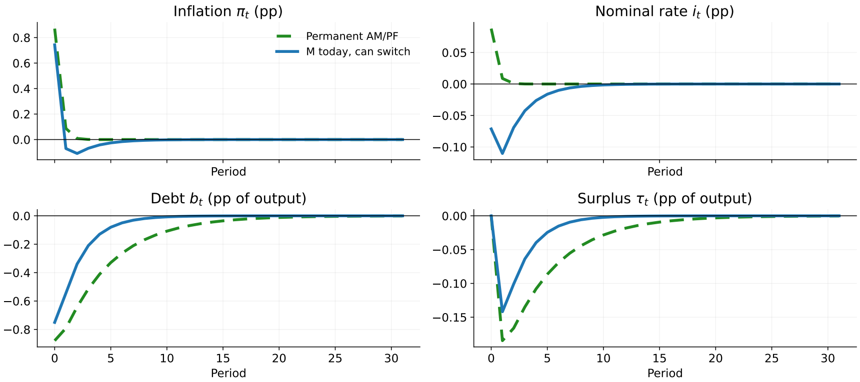
IRF: FISCAL SHOCK WITH SWITCHING RISK



Units: π_t, i_t in pp; b_t, τ_t in pp of output. Shock: $\psi_0 = -1$. Calibration: $\beta = .99$, $\rho_\theta = .10$, $\rho_\psi = .70$, $p_{MF} = .10$, $p_{FF} = .85$.

IRF: MONETARY RATE-INCREASE SHOCK WITH SWITCHING RISK

Regime M today with p(M to F)=0.10: monetary rate-increase shock



Units: π_t, i_t in percentage points; b_t, τ_t in points of output. Shock: $\theta_0 = -1$; raises i_t in M.
Calib: $\beta = 0.99$, $\rho_\theta = 0.10$, $\rho_\psi = 0.70$; M $\phi_\pi = 1.25$, $\lambda_b = 0.80$; F $\phi_\pi = 0.00$, $\gamma = 0.00$; $p_{MF} = 0.10$, $p_{FF} = 0.85$.

Units: π_t, i_t in pp; b_t, τ_t in pp of output. Shock: $\theta_0 = -1$; this raises i_t in Regime M. Calibration: $\beta = .99$, $\rho_\theta = .10$, $\rho_\psi = .70$, $p_{MF} = .10$, $p_{FF} = .85$.

TAYLOR COEFFICIENTS CHANGE VALUATION COEFFICIENTS

In the simple switching model, a stronger Taylor response changes the equilibrium valuation mapping:

$$g_i^{\psi} = -\frac{H_i}{\beta\phi_{\pi,i} + \beta^{-1}H_i}, \qquad H_i \equiv \sum_j p_{ij}h_j.$$

If agents assign positive probability to active fiscal policy, the current fiscal-shock coefficient depends on both today's monetary feedback and future Regime-F valuation terms; $\partial g_M^{\psi}/\partial\phi_{\pi}(M)$ can be positive.

- The mechanism is coefficient selection across regimes, not an IS-curve demand channel.
- Active monetary policy today need not insulate inflation from fiscal shocks when future fiscal backing may be active.

IDENTIFICATION: REGIMES ARE NOT OBSERVED DIRECTLY

- Cochrane (1999), Leeper–Walker (2013): in simple linear settings, the same observables can often be rationalized under either regime.
- Reason: with suitable policy-rule and shock specifications, the same state vector

$$X_t \equiv (\epsilon_t^M, \epsilon_t^F, \hat{b}_{t-1})$$

  can support different structural rules.
- The same equilibrium mapping $i_t = f(X_t)$ may be generated by different policy rules because structural rules restrict off-equilibrium behavior, not only equilibrium correlations.
- Simple correlations cannot distinguish the regimes.

TAYLOR-RULE OLS TRAP: WHAT OLS ESTIMATES

- Structural rule and monetary shock:

$$i_t = \phi_\pi \pi_t + \theta_t, \qquad \theta_t = \rho_\theta \theta_{t-1} + \varepsilon_t^\theta.$$

- Equilibrium data obey:

$$i_t = f(\theta_t), \qquad \pi_t = g(\theta_t).$$

- OLS estimates

$$\hat{\delta} = \frac{\mathrm{Cov}(i_t, \pi_t)}{\mathrm{Var}(\pi_t)},$$

not the structural Taylor coefficient ϕ_π.

CORRECT IDENTIFICATION STRATEGY

- Impose the bond valuation equation as a cross-equation restriction.
- Estimate monetary and fiscal rules jointly (Chung–Leeper 2007).
- Use a fully specified DSGE model with both regimes (Traum–Yang; Leeper–Traum–Walker).
- Allow Markov-switching coefficients (Bianchi; Bianchi–Ilut; Chen–Leeper–Leith).
- Reduced-form correlations do not suffice on their own.

VALUATION WITH LOW RATES AND DEFAULT

BASELINE FTPL VALUATION WITH AN SDF

Maintained assumption: promised nominal payments are always honored.

$$\frac{R_{t-1}B_{t-1}}{P_t} = \mathbb{E}_t \sum_{j=0}^{\infty} M_{t,t+j}\tau_{t+j}.$$

- Left side: real value of inherited nominal promises.
- Right side: risk-adjusted present value of primary surpluses.
- If the present-value operator converges and debt is default-free, the equation can select P_t.
- The next two modifications ask what happens when discounting is weak or repayment is defaultable.

LOW REAL RATES CHALLENGE PRICE-LEVEL SELECTION

- Across advanced economies, real government-debt returns have often been below growth rates for sustained periods (Bassetto–Cui 2018).
- The simple PV version of the FTPL loses clean selection power when the relevant present-value operator fails to converge.
- Low government-debt returns can come from risk adjustment, dynamic inefficiency, or liquidity services.
- Therefore the implication for price-level selection depends on why rates are low.

BASSETTO–CUI: DEBT LAW

Use compact debt-output notation:

$$b_t^Y \equiv \frac{B_t}{P_t Y_t}, \qquad x_t \equiv \frac{\tau_t}{Y_t}, \qquad m_{t+1} \equiv \frac{1 + r_{t+1}}{1 + g_{t+1}}.$$

The government budget is

$$b_{t+1}^Y = m_{t+1}(b_t^Y - x_t),$$

where τ_t is a real primary surplus and $x_t > 0$ is a surplus-output ratio.

- Low government-debt returns mean $m_{t+1} < 1$ for sustained periods.
- Bassetto–Cui ask whether the fiscal valuation equation still selects the price level.

BASSETTO–CUI: ITERATION

Iterating

$$b_{t+1}^{Y} = m_{t+1}(b_t^{Y} - x_t)$$

from date 0 gives

$$b_t^{Y} = b_0^{Y} \prod_{v=1}^{t} m_v - \sum_{s=0}^{t-1} x_s \prod_{v=s+1}^{t} m_v.$$

If $m_v \leq \alpha < 1$ persistently,

$$b_0^{Y} \prod_{v=1}^{t} m_v \leq \alpha^t b_0^{Y} \to 0.$$

Thus positive debt bounded away from zero requires the weighted future surplus term to be negative.

BASSETTO–CUI: RISK-ADJUSTED DISCOUNTING

In the stochastic representative-agent case, define the stochastic discount factor

$$M_{t,t+1} = \beta \frac{u'(c_{t+1})}{u'(c_t)}.$$

The bond Euler equation is

$$1 = \mathbb{E}_t \left[M_{t,t+1} (1 + R_t) \frac{P_t}{P_{t+1}} \right],$$

- The government-debt return can be low in expectation.
- But the relevant FTPL object is the risk-adjusted discount factor.

RISK PREMIA: LOW EXPECTED RETURNS NEED NOT BREAK FTPL

Household transversality gives

$$\frac{B_t}{P_t} = \mathbb{E}_t \sum_{s=0}^{\infty} M_{t,t+s}\tau_{t+s}.$$

The raw expected debt return may be below growth:

$$\mathbb{E}_t \left[\frac{(1+R_t)P_t Y_t}{P_{t+1}Y_{t+1}} \right] < 1.$$

Even so, the risk-adjusted PV can be well-defined. Expected primary surpluses can be negative while their risk-adjusted PV is positive.

DYNAMIC INEFFICIENCY: DEBT CAN BE MONEY-LIKE

In the OLG example, young households' real saving s_t pins down debt demand. The equilibrium recursion is

$$s_{t+1} = (1 + r(s_t))\, s_t - \tau, \qquad s_1 = \frac{B_0}{P_0} - \tau.$$

Dynamic inefficiency means households want to save even at a zero real return:

$$r(0) < 0.$$

If τ is not too negative, many initial savings levels are consistent with equilibrium:

$$s_1 \in (-\infty, \bar{s}^{\max}].$$

Since $B_0/P_0 = \tau + s_1$, a continuum of s_1 implies a continuum of admissible P_0.

LIQUIDITY SERVICES: DEBT DEMAND CONSTRAINT

With liquidity services, government debt earns a low return because it relaxes a constraint or facilitates trade. The law of motion is similar:

$$s_{t+1} = (1 + r(s_t))\, s_t - \tau,$$

but debt positions must satisfy

$$s_t \geq 0.$$

This constraint changes the selection problem relative to the dynamically inefficient OLG case.

LIQUIDITY SERVICES: WHEN DOES FTPL RETURN?

Bassetto–Cui's selection result is conditional:

$$\tau > 0 \Rightarrow s_1 = \bar{s} \Rightarrow \frac{B_0}{P_0} = \tau + \bar{s} \quad \text{unique } P_0,$$

but then the steady-state return satisfies $r(\bar{s}) > 0$ in their zero-growth normalization.

- Low debt returns with positive debt point toward $\tau \leq 0$ eventually.
- With $\tau \leq 0$, multiple price levels reappear.

LOW-RATE MECHANISMS AND PV CONVERGENCE

- Risk-adjusted discounting: raw expected returns can be low while the risk-adjusted PV still converges.
- Dynamic inefficiency: government debt can be money-like, and P_t selection may fail.
- Liquidity services: selection depends on debt demand constraints and the surplus path τ_t.
- Bassetto–Cui (2018): low government-debt returns are not one case; they are several mechanisms.
- Even when not unique, the fiscal valuation condition can still provide a lower bound on the price level.

BASELINE FTPL ASSUMES DEFAULT-FREE NOMINAL DEBT

- Maintained assumption: promised nominal payments are always honored.

$$\frac{R_{t-1}B_{t-1}}{P_t} = \mathbb{E}_t \sum_{j=0}^{\infty} M_{t,t+j}\tau_{t+j}.$$

- If the present value of surpluses falls, the price level jumps so that the real repayment value of nominal promises falls.
- This is not legal default.
- It is real repayment dilution through the price level.

DEFAULTABLE DEBT ADDS AN ADJUSTMENT MARGIN

- Let δ_t be Uribe's default/subsidy rate and $\rho_t = 1 - \delta_t$ the repayment rate.
- The valuation condition becomes

$$\rho_t \frac{R_{t-1}B_{t-1}}{P_t} = \mathbb{E}_t \sum_{j=0}^{\infty} M_{t,t+j}\tau_{t+j}.$$

- One valuation equation, two adjustment margins: δ_t and P_t.
- Conditional on a price level P_t,

$$\delta_t(P_t) = 1 - \frac{\mathbb{E}_t \sum_{j=0}^{\infty} M_{t,t+j}\tau_{t+j}}{R_{t-1}B_{t-1}/P_t}.$$

- This pins down real repayment, not the split between inflation and default.
- In Uribe's analytical benchmark δ_t can be negative; a literal haircut would add $\delta_t \geq 0$.

BOND PRICING WITH DEFAULT RISK

The one-period nominal bond satisfies

$$1 = \mathbb{E}_t \left[M_{t,t+1} R_t (1 - \delta_{t+1}) \frac{P_t}{P_{t+1}} \right].$$

- With constant discounting, $M_{t,t+1} = \beta$, giving Uribe's simplified expression.
- Default risk is not outside the FTPL logic; it enters the asset-pricing problem.
- Monetary policy affects whether fiscal stress is forecastable and therefore priced at issuance.

URIBE SETUP: DEFAULTABLE ONE-PERIOD DEBT

Constant endowment implies a constant real SDF. Uribe's government budget is

$$B_t = R_{t-1}B_{t-1}(1 - \delta_t) - P_t\tau_t.$$

- B_t: nominal debt issued at date t.
- $R_{t-1}B_{t-1}$: promised nominal repayment inherited from last period.
- δ_t: default/subsidy rate on inherited liabilities.
- τ_t: real primary surplus.

URIBE VALUATION: REPAYMENT VALUE

Iterating the government budget forward gives

$$\frac{(1-\delta_t)R_{t-1}B_{t-1}}{P_t} = \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j \tau_{t+j} \equiv \mathrm{PV}_t(\tau).$$

- The valuation equation prices what bondholders actually receive.
- Baseline FTPL is the special case $\delta_t = 0$.

URIBE VALUATION: DEFAULT/SUBSIDY RATE

Conditional on a price level P_t,

$$\delta_t(P_t) = 1 - \frac{\mathrm{PV}_t(\tau)}{R_{t-1}B_{t-1}/P_t}.$$

If $\delta_t = 0$, the price level must satisfy

$$\frac{R_{t-1}B_{t-1}}{P_t} = \mathrm{PV}_t(\tau).$$

- With default, the equation pins down repayment value.
- Additional policy structure determines whether P_t, δ_t, or both adjust.

CURRENT-LOOKING TAYLOR RULE

Let $\Pi_t \equiv P_t / P_{t-1}$ denote gross inflation in this Uribe block. The current-looking Taylor rule is

$$R_t = R^* + \phi_\pi(\Pi_t - \Pi^*), \qquad R^* = \frac{\Pi^*}{\beta}.$$

- Active monetary policy means $\phi_\pi \beta > 1$.
- The rule reacts to current inflation, not expected future inflation.

CURRENT-LOOKING RULE: BOUNDED INFLATION SELECTION

If default is ruled out, the one-period bond is riskless:

$$1 = \beta R_t \frac{1}{\Pi_{t+1}} \quad \Rightarrow \quad \Pi_{t+1} = \beta R_t.$$

Substituting the current-looking rule:

$$\Pi_{t+1} = \Pi^* + \phi_\pi \beta (\Pi_t - \Pi^*).$$

Equivalently,

$$\Pi_{t+1} - \Pi^* = \phi_\pi \beta (\Pi_t - \Pi^*).$$

- With $\phi_\pi \beta > 1$, boundedness selects $\Pi_t = \Pi^*$ for all t.
- This is the monetary selection result; it does not yet impose fiscal backing.

NO-DEFAULT EQUILIBRIUM REQUIRES FISCAL CONSISTENCY

Under $\delta_t = 0$, fiscal valuation requires

$$\frac{R_{t-1}B_{t-1}}{P_t} = \mathrm{PV}_t(\tau).$$

For a deterministic surplus path,

$$\mathrm{PV}_t(\tau) = \sum_{j=0}^{\infty} \beta^j \tau_{t+j}; \qquad \tau_{t+j} = \bar{\tau} \Rightarrow \mathrm{PV}_t(\tau) = \frac{\bar{\tau}}{1-\beta}.$$

NO-DEFAULT SELECTION PINS DOWN INFLATION

The bounded current-looking Taylor rule selects target gross inflation:

$$\Pi_t = \Pi^*.$$

Conditional on inherited P_{t-1}, this implies the continuation price path $P_t = \Pi^* P_{t-1}$. Therefore a bounded no-default equilibrium exists only if

$$\frac{R_{t-1}B_{t-1}}{\Pi^* P_{t-1}} = \mathrm{PV}_t(\tau).$$

- Constant τ_t is only the steady-state benchmark.
- Uribe’s default mechanism needs fiscal news that moves $\mathrm{PV}_t(\tau)$.
- If that breaks the equality, another margin must move: δ_t, P_t, or future τ_t.

CURRENT RULE WITH PRICE STABILITY: EXPECTED DEFAULT

Suppose the central bank keeps $\Pi_t = \Pi^*$, so $R_t = \Pi^*/\beta$. Bond pricing becomes

$$1 = \beta \frac{\Pi^*}{\beta} \mathbb{E}_t \left[(1 - \delta_{t+1}) \frac{1}{\Pi^*} \right] = \mathbb{E}_t (1 - \delta_{t+1}).$$

Therefore

$$\mathbb{E}_t \, \delta_{t+1} = 0.$$

- In Uribe's analytical benchmark, δ_t can be negative; negative default is a subsidy to bondholders.
- A literal nonnegative haircut would require an additional $\delta_t \geq 0$ constraint.
- Under the unconstrained convention, default/subsidy is not forecastable at issuance.

CURRENT RULE WITH PRICE STABILITY: REALIZED DEFAULT

Under the target path and one-period debt,

$$\mathrm{PV}_{t|q} \equiv \sum_{h=0}^{\infty} \beta^h \, \mathbb{E}_q \, \tau_{t+h}.$$

The realized default/subsidy rate is

$$\delta_t = 1 - \frac{\mathrm{PV}_{t|t}}{\mathrm{PV}_{t|t-1}}, \qquad t \geq 1.$$

- Default occurs when fiscal news lowers the PV of future surpluses.
- The current-looking Taylor rule turns fiscal news into surprise default.
- One-period spreads need not move before the news arrives.

FORWARD-LOOKING TAYLOR RULE

Uribe's forward-looking rule reacts to expected inverse inflation:

$$R_t = R^* + \phi_\pi \left(\frac{1}{\mathbb{E}_t(P_t/P_{t+1})} - \Pi^* \right).$$

With $\delta_t = 0$, there is an equilibrium with

$$R_t = \frac{\Pi^*}{\beta}, \qquad \frac{1}{\mathbb{E}_t(P_t/P_{t+1})} = \Pi^*.$$

- Expected inflation is anchored by the policy rule.
- The timing change matters for whether default is needed.

FORWARD-LOOKING RULE: FISCAL NEWS

The valuation equation still holds:

$$\frac{R_{t-1}B_{t-1}}{P_t} = \mathrm{PV}_t(\tau).$$

Fiscal news moves the realized price level through $\mathrm{PV}_t(\tau)$.

- Realized inflation can deviate from target.
- The deviations are unforecastable under the rule.
- Default is avoidable in the benchmark.

PRICE-LEVEL PEG: VALUATION

Under a price-level peg, $P_t = 1$, so the risk-free nominal rate is

$$R_t^f = \beta^{-1}.$$

The valuation equation becomes

$$\delta_t = 1 - \frac{\mathrm{PV}_t(\tau)}{R_{t-1} B_{t-1}}.$$

The equilibrium debt stock satisfies

$$B_t = \sum_{h=1}^{\infty} \beta^h \, \mathbb{E}_t \, \tau_{t+h}.$$

PRICE-LEVEL PEG: FORECASTABLE SPREADS

Bond pricing under the peg gives

$$1 = \beta R_t \, \mathbb{E}_t(1 - \delta_{t+1}), \qquad \frac{R_t}{R_t^f} = \frac{1}{\mathbb{E}_t(1 - \delta_{t+1})}.$$

Persistent fiscal weakness lowers expected repayment:

$$\mathbb{E}_t(1 - \delta_{t+1}) \downarrow \quad \Rightarrow \quad R_t / R_t^f \uparrow.$$

- Under a peg, bad fiscal news cannot be absorbed by inflation.
- Default can become persistent and priced ex ante.

URIBE TIMING: THREE CASES

Uribe-style timing model: one-period nominal debt, active fiscal policy, and fiscal news shocks.

- Current Taylor rule: fiscal-news default is a surprise; no pre-news spread in the benchmark.
- Forward Taylor rule: expected inflation is anchored; default is avoidable in the benchmark.
- Price-level peg: default is persistent and forecastable; sovereign spreads can persist.

Small timing differences in the monetary rule change the equilibrium distribution of default.

DELAYED DEFAULT: PRICE-LEVEL PATH

Use a price-level ratio relative to the target path:

$$p_t \equiv \frac{P_t}{P_t^*}.$$

If fiscal imbalances imply $p_0 > 1$ and default is delayed:

$$p_t = 1 + (\phi_\pi \beta)^t (p_0 - 1).$$

- Perfect-foresight transition: no default for $0 \leq t < T$.
- At T, default restores the target price path.
- With $\phi_\pi \beta > 1$, the price-level gap grows over time.
- Temporary delay does not remove the fiscal imbalance.

DELAYED DEFAULT: TERMINAL ADJUSTMENT

If default occurs at date T, the default rate required to restore price stability is

$$\delta_T = 1 - \frac{1}{p_T} = 1 - \frac{1}{1 + (\phi_\pi \beta)^T (p_0 - 1)}.$$

- Waiting raises both the interim price-level gap and the eventual default.
- Real debt need not warn policymakers: in Uribe's transition, real public debt can stay flat until the crisis.

FISCAL COMPATIBILITY

FISCAL COMPATIBILITY OF INFLATION TARGETING

Given inherited nominal liabilities and target price path P_t^*:

$$\frac{R_{t-1}B_{t-1}}{P_t^*} = \mathbb{E}_t \sum_{j=0}^{\infty} M_{t,t+j}\tau_{t+j}.$$

- Exact inflation targeting imposes a restriction on fiscal backing.
- Too-low fiscal backing $\Rightarrow$ higher P_t, default, or future fiscal correction.
- Too-high fiscal backing $\Rightarrow$ target path may require fiscal relaxation or other valuation adjustment.
- Inflation targeting is a joint monetary–fiscal commitment.

FISCAL RULES AS BACKING RESTRICTIONS

- Sweden: net lending target is 1/3% of GDP over the cycle through 2026; balance target from 2027.
- Switzerland: debt brake on expenditure growth.
- Germany: debt brake, with 0.35% structural-deficit cap and post-2025 exemptions / reform debates.
- UK: current budget targeted to be in surplus by 2029–30, plus debt-related rules.
- Most fiscal rules are framed as deficit, expenditure, or balance constraints.
- Debt anchors, where present, are often benchmarks rather than automatic state-contingent relaxation rules.
- Sustained austerity can make $\mathbb{E}_t \sum_j M_{t,t+j}\tau_{t+j}$ exceed what is compatible with P_t^*.

INTEREST-RATE NORMALIZATION CHANGES THE BACKING REQUIREMENT

- Post-crisis: debt-to-GDP doubled in most advanced economies.
- Normalization raises debt service for a given debt stock.
- CBO projects US net interest spending at 3.3% of GDP in 2026 and 4.6% in 2036.
- Those costs must be financed by higher taxes, lower spending, faster debt growth, default, or inflation.
- Higher debt raises the temptation to inflate $\Rightarrow$ time-inconsistency problem (Leeper–Leith–Liu).
- King (1995): disinflation can itself destabilize debt.

CONSOLIDATED GOVERNMENT BUDGET CONSTRAINT

A compact consolidated-government version is

$$\frac{B_{t-1} + M_{t-1} - A_{t-1}}{P_t} = \mathbb{E}_t \sum_{j=0}^{\infty} M_{t,t+j} \left(\tau_{t+j} + \text{seigniorage}_{t+j} \right).$$

- Monetary and fiscal liabilities are jointly backed by fiscal resources, seigniorage, and assets.
- Open-market operations affect prices only if they change expected backing, liquidity services, risk, or constraints.
- Otherwise they are liability-composition swaps.

SUMMARY

MAIN RESULTS: REGIMES

- In the fixed-regime benchmark, determinacy requires one unstable root because there is one jump variable and one predetermined state.
- In Regime M, $g_M^\psi = 0$: active monetary / passive fiscal policy insulates inflation from fiscal shocks.
- In Regime F, $g_F^\psi \neq 0$: fiscal shocks move inflation because the price level revalues inherited nominal debt.
- Under Markov switching, equilibrium coefficients depend on the transition matrix Π, not only on today's policy coefficients.
- Reduced-form Taylor and surplus-debt regressions estimate equilibrium comovements, not structural regime parameters.

MAIN RESULTS: VALUATION

- Low government-debt returns do not by themselves determine FTPL uniqueness; the relevant object is the risk-adjusted PV operator.
- With defaultable debt, the valuation equation pins down real repayment, not the split between inflation and default.
- Inflation targeting requires fiscal backing consistent with the target price path.

APPENDIX

APPLICATION OF IDENTIFICATION CAVEAT: HISTORICAL REGIME LABELS

Period	Interpretation	Source
1942–1951	F (Fed pegged 3/8% T-bills)	Woodford (2001)
1960s–1979	F-like (weak Taylor response plus fiscal evidence)	CGG; Bianchi–Ilut
1980–2007	M-like, with switches	Chen–Leeper–Leith
2008–2015; 2020–2022	Ambiguous at ZLB	Open question
2022–2026	High-rate monetary tightening; fiscal side disputed	Open question
Brazil 2015	Loyo-style AM/AF interpretation	Leeper–Leith; Economist

- Regime labels require joint monetary–fiscal evidence, not a Taylor coefficient alone.

EMPIRICAL IMPLICATION: PASSIVE-FISCAL EXPECTATIONS CAN REVERSE INFLATION RESPONSES

- Stylized VAR fact: monetary easing is sometimes followed by lower measured inflation.
- Standard explanations include cost channels and omitted information.
- Switching explanation: in Regime F, or with probability of a switch to F, lower i_t today lowers expected future debt service.
- Lower expected fiscal stress can lower expected future inflation, hence lower current inflation.
- This is one structural explanation of price-puzzle responses, not a universal account.

THE 2008–2014 PUZZLE RECONSIDERED

- Federal debt rose sharply while the federal funds target was constrained near zero.
- Naive FTPL reading: higher nominal debt should imply much higher inflation.
- Where is the inflation?
- One reconciliation is that lower real discount rates raised the present value of expected surpluses, offsetting the increase in nominal debt in the valuation equation.
- B_{t-1}/P_t does not have to rise. The richer FTPL is consistent with stable inflation.
- Simple “debt up $\Rightarrow$ inflation up” is a wrong reading of the theory.

WALLACE IRRELEVANCE THEOREM

- At an interest-on-reserves Friedman rule ($R_t^M = R_t^B$), money and bonds are perfect substitutes.
- An open-market swap of money for bonds is then irrelevant:

$$\Delta M_t = -\Delta B_t \quad \Rightarrow \quad \text{no change in } P_t, c_t, R_t.$$

- Wallace (1981), Chamley–Polemarchakis (1984): Modigliani–Miller in monetary economies.
- Open-market operations matter when they change fiscal backing, liquidity services, risk, or constraints.

MODEL F VS MODEL E CENTRAL BANKS (SIMS 2004)

Model F (Fed, BoE)

- CB holds only government bonds.
- Positive seigniorage flows to Treasury.
- CB assets are purely nominal.
- Needs Treasury fiscal backstop.

Model E (ECB, HKMA)

- CB holds real assets: gold, FX reserves.
- Fiscal backstop partly in-house.
- Natural for currency unions with heterogeneous fiscal authorities.
- FX reserves push the backing problem one level out.

- Persistent negative net worth requires fiscal backing, future seigniorage, valuation gains, or institutional tolerance.

SARGENT 1982: HYPERINFLATION ENDINGS AS FISCAL REGIME CHANGES

- Studied post-WWI hyperinflations in Austria, Hungary, Germany, Poland.
- All four ended abruptly after policy changes.
- Common pattern:
  - credible commitment to a balanced budget;
  - independence of the central bank from fiscal authority;
  - sometimes external conditionality.
- The historical interpretation is fiscal regime change, not monetary discipline alone.

BRAZIL 2015 AS A LOYO-STYLE AM/AF INTERPRETATION

- Constitutional benefit indexation and mandatory spending rules make roughly 90% of expenditures rigid or outside ordinary legislative control.
- 2015 overall public-sector deficit: about 10.3% of GDP; primary deficit: about 1.9% of GDP.
- Banco Central do Brasil raised policy rate to 14.25%.
- IPCA inflation reached 10.67% in 2015.
- Mechanism: higher i_t raises debt service; if fiscal policy cannot offset, expected inflationary/default adjustment rises.

BRAZIL 2016–2023: FISCAL RULE ADOPTION AND EROSION

- December 2016: Constitutional spending cap (*Teto de Gastos*) limited federal primary spending growth to previous-year inflation for 20 years.
- Interpretation: institutional commitment toward passive fiscal policy.
- Inflation: 10.7% (2015) → 2.9% (2017). Selic cut from 14.25% to 6.5% by 2018.
- 2020–: COVID + political pressure effectively breach the cap.
- 2023: Lula’s government replaces the cap with a softer “Marco Fiscal Sustentável”.
- Lesson: fiscal-regime commitments can weaken over time.

ARGENTINA 2024: FISCAL CONSOLIDATION AS REGIME CHANGE

- Dec 2023: Milei takes office. Annual CPI inflation ~211%.
- Inherited: large primary deficits monetized by the BCRA.
- IMF description: 2024 fiscal adjustment around 5% of GDP.
- BCRA stops financing the Treasury; immediate currency devaluation.
- Monthly inflation: 25.5% (Dec 2023) → 2.7% (Dec 2024).
- 2024: primary surplus about 1.8% of GDP; overall cash surplus about 0.3% of GDP.
- Fiscal consolidation was central, but devaluation, recessionary compression, administered prices, and capital controls also mattered.

OPEN QUESTIONS

- How do we identify the regime in real time, before history settles it?
- How should sovereign default risk enter regime classification? (Open economy, monetary unions.)
- Can fiscal rules be redesigned to be compatible with an inflation target?
- What role for political economy in choosing which authority becomes passive?
- Does central-bank balance-sheet policy blur the regime classification entirely?